Notes on Lattices

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Abstract

Notes taken while studying lattice-based cryptography.

Usually while reading books and papers I take handwritten notes in paper notebooks, this document contains some of them re-written to $LaTeX$.

Contents

1 Lattices

1.1 Hard problems

Definition 1.1 (SVP (Shortest Vector Problem))**.** find a shortest non-zero vector $x \in \mathcal{L}$, ie. that minimizes the Euclidean norm $||x||$, that is $\lambda_1(\mathcal{L}) = ||x||$.

Definition 1.2 (CVP (Closest Vector Problem))**.** given a vector $y \in \mathbb{R}^n$ with $y \notin \mathcal{L}$, find a nonzero vector $x \in \mathcal{L}$ that is closest to y, ie. that minimizes the Euclidean norm $||y - x||$.

Definition 1.3 (NTRU key recovery problem)**.** Given $h(x)$, find ternary polynomials $f, g \in \mathcal{T}$ satisfying

$$f(x) \cdot h(x) = g(x) \pmod q \in R_q$$

Where $R_q = \mathbb{Z}_q[X]/(X^n - 1)$.

Definition 1.4 (SIS (Short Integer Solution))**.** Given $q \in \mathbb{Z}$ and matrix $A \in \mathbb{Z}_q^{m \times n}$, find a non-zero short $z \in \mathbb{Z}^n$ s.th.

$$A \cdot z = 0 \pmod q$$

"short": $z \in \{B, B\}^n$ for some bound B, often $z \in \{-1, 0, 1\}^n$.

The set of all solutions to the SIS problem forms the (q-ary) lattice

$$\Lambda_q^{\perp}(A) = \{z \in \mathbb{Z}^n \mid Az = 0 \in \mathbb{Z}_q^m\} \supseteq \mathbb{Z}_q^n$$

where the SIS problem becomes the SVP for $\Lambda_q^{\perp}(A)$

- LLL algorithm is used to solve SVP and CVP.
- Both SVP and CVP become computationally difficult (NP hard) as the lattice dimension n grows.

Definition 1.5 (Ring SIS (Ring Short Integer Solution))**.** given a uniformly random matrix $A \in^R R_q^n$, with elements $a_i \in R_q$, find a non-zero integer vector $z \in R^n$ of norm $||z|| \leq \beta$ such that

$$Az = \sum_i a_i \cdot z_i = 0 \in R_q$$

Definition 1.6 (MSIS (Module Short Integer Solution))**.** given a uniformly random matrix $A \in^R R_q^{m \times n}$, with columns $a_i \in R_q^m$, find a non-zero integer vector $z \in R^n$ of norm $||z|| \leq \beta$ such that

$$Az = 0 \in R_q^m$$
$$ie. \sum_i a_{ij} \cdot z_j = 0 \in R_q, \quad for\ i = 1, \ldots, m$$

Notice that Module-SIS is a Ring-SIS with $rank = 1$ ($m = 1$).

The next form is useful to compress the size of a SIS matrix by n columns:

Definition 1.7 (Hermite Normal Form (HNF))**.** Assume the leftmost n columns $A_1 \in \mathbb{Z}_q^{n \times n}$ of $A = [A_1|A_2] \in \mathbb{Z}_q^{n \times m}$ form an invertible matrix.

Then replace A by

$$A_1^{-1} \cdot A = [I_n \mid \overline{A} = A_1^{-1} \cdot A_2]$$

and threat the I_n submatrix as implicit.

Then, A and $[I_n \mid \overline{A}]$ have exactly the same set of short SIS solutions.

Definition 1.8 (LWE distribution)**.** $s \in \mathbb{Z}_q^n$ (secret), $a \in \mathbb{Z}_q^n$, $e \in \Xi$, output

$$(a, b = \langle s, a \rangle + e \pmod q) \in (\mathbb{Z}_q^n \times \mathbb{Z}_q)$$

Definition 1.9 (Search-LWE)**.** given m independent samples $(a_i, b_i) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$, for a uniform $s \in \mathbb{Z}_q^n$ (fixed for all samples), find s.

Definition 1.10 (Decision-LWE)**.** given m independent samples $(a_i, b_i) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$, distinguish which one is not random and it is an actual instance.

1.2 Dual lattices

From previous section, we have LWE-Lattice and SIS-Lattice:

Search-LWE can be seen as a Bounded-Distance Decoding (BDD) problem on q-ary m-dimensional integer lattices:
for LWE samples, vector b is relatively close to exactly one vector in the LWE lattice:

Definition 1.11 (LWE Lattice)**.** A LWE-Lattice $\mathcal{L}$ is defined

$$\begin{aligned}\mathcal{L} &= \{A^T s \mid s \in \mathbb{Z}_q^n\} + q\mathbb{Z}^m \\ &= \{A^T s + qz \mid s \in \mathbb{Z}_q^n, z \in \mathbb{Z}^m\}\end{aligned}$$

SIS can be seen as a SVP on q-ary m-dimensional lattices:

Definition 1.12 (SIS-Lattice)**.** A SIS-Lattice $\mathcal{L}$ is defined

$$\mathcal{L}^{\perp} = \{z \in \mathbb{Z}^m \mid Az = 0 \in \mathbb{Z}_q^n\} \supseteq q\mathbb{Z}^m$$

Definition 1.13 (Dual of a lattice)**.** For a lattice $\mathcal{L}$, its dual is

$$\mathcal{L}^* = \{y \in \mathbb{R}^m \mid \langle y, x\rangle \in \mathbb{Z} \ \forall \ x \in \mathcal{L}\}$$

The dual lattice consists of all vectors that measure lattice points with integer-valued linear measurements.

Remark 1.14. For a lattice basis $B = (b_1, \ldots, b_n)$, the dual basis $B^* = (b_1^*, \ldots, b_n^*)$ is the unique basis satisfying

$$\langle b_i^*, b_j\rangle = \delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j \end{cases}$$

where δ_{ij} is the Kronecker delta.

Lemma 1.15.

$$\mathcal{L} = q\mathcal{L}^{\perp *} \text{ ie. } \frac{1}{q}\mathcal{L} = \mathcal{L}^{\perp *}$$

Proof. i. Take $y \in \frac{1}{q}\mathcal{L}$, then by 1.11, $\exists \ s \in \mathbb{Z}^n, z \in \mathbb{Z}^m$ such that

$$y = \frac{1}{q}(A^T s + qz) = \frac{1}{q}A^T s + z$$

Let $x \in \mathcal{L}^{\perp}$. By definition (1.12) $Ax = 0 \pmod q$, therefore $\exists \ k \in \mathbb{Z}^n$ such that $Ax = qk$.

Compute

$$\begin{aligned}
\langle y, x\rangle &= \langle \frac{1}{q}A^T s + z, x\rangle \\
&= \frac{1}{q}\langle A^T s, x\rangle + \langle z, x\rangle \\
&= \frac{1}{q}\langle s, Ax\rangle + \langle z, x\rangle \\
&= \frac{1}{q}\langle s, qk\rangle + \langle z, x\rangle \\
&= \langle s, k\rangle + \langle z, x\rangle
\end{aligned}$$

and $\langle s, k\rangle + \langle z, x\rangle \in \mathbb{Z}$, since each of $s, k, z, x \in \mathbb{Z}$ separately.

Thus $\langle y, x\rangle \in \mathbb{Z}$, hence by 1.13, $y \in \mathcal{L}^{\perp *}$.

We started from $y \in \frac{1}{q}\mathcal{L}$, so, now we have

$$y \in \mathcal{L}^{\perp *} \ \forall\ y \in \frac{1}{q}\mathcal{L}$$

therefore

$$\mathcal{L}^{\perp *} \supseteq \frac{1}{q}\mathcal{L}$$

ii. Now we prove the reverse inclusion. Take $y \in \mathcal{L}^{\perp *}$.

Observe that $q\mathbb{Z}^m \subseteq \mathcal{L}^{\perp}$, because

$$\forall z \in \mathbb{Z}^m,\ A(qz) = qAz = q0 = 0 \pmod q$$

Since $y \in \mathcal{L}^{\perp *}$, we must have $\langle y, qz\rangle \in \mathbb{Z}\ \forall\ z \in \mathbb{Z}^m$.

Let $e_i \in \mathbb{Z}^m$ be the $i-th$ standard basis vector, ie. $e_i = (0, \ldots, 0, 1, 0, \ldots, 0)$ (where one is at the i-th position).

We know that $\langle y, qz\rangle \in \mathbb{Z}\ \forall z \in \mathbb{Z}^m$ (including $z = e_i$), thus, for $qe_i = (0, \ldots, 0, q, 0, \ldots, 0)$,

$$\langle y, qe_i\rangle = q\langle y, e_i\rangle = q \cdot y_i$$

since $\langle y, e_i\rangle = y_i$ (extracts the i-th position from y).

So, since $\langle y, qz\rangle \in \mathbb{Z}\ \forall z \in \mathbb{Z}^m$, and $\langle y, qe_i\rangle = qy_i$, then $qy_i \in \mathbb{Z}$

This argument works $\forall i \in [m]$, that is, $qy_1, \ldots, qy_m \in \mathbb{Z}$. Therefore

$$qy \in \mathbb{Z}^m\ \forall\ y \in \mathcal{L}^{\perp *}$$

For $\overline{x} \in ker(A) \subseteq \mathbb{Z}_q^m$, since $A\overline{x} = 0$, then $Ax = 0 \pmod q$, thus $x \in \mathcal{L}^{\perp}$ (by its definition 1.12).

Since $y \in \mathcal{L}^{\perp *}$, by the definition of $\mathcal{L}^{\perp *}$ (1.13), $\langle y, x\rangle \in \mathbb{Z}$. Multiply it by q,

$$q\langle y, x\rangle = \langle qy, x\rangle \in q\mathbb{Z}$$

hence $\langle qy, x\rangle = 0 \pmod q$, thus $qy \pmod q \in ker(A)^{\perp}$.

We'll now prove the following identity, valid for every integer $q \geq 2$, which we will use later:

$$(\ker A)^{\perp} = im(A^T)$$

where $A : \mathbb{Z}_q^m \to \mathbb{Z}_q^n$.

Let $y \in im(A^T)$. Then $\exists\ s \in \mathbb{Z}_q^n$ such that $y = A^T s$.

Then, $\forall\ x \in \ker(A)$, by definition of the kernel, $Ax = 0$. Want to prove that y is orthogonal to every $x \in \ker(A)$:

$$\langle y, x \rangle = \langle A^T s, x \rangle = \langle s, Ax \rangle = 0$$

thus, $\langle y, x \rangle = 0\ \forall\ x \in \ker(A)$, hence y is orthogonal to every x, that is, $y \in (\ker(A))^{\perp}$.

Therefore,

$$im(A^T) \subseteq (\ker(A))^{\perp}$$

For the reverse inclusion, it suffices to show that both sets have the same number of elements:

By the First Isomorphism Theorem,

$$|im(A)| = \frac{|\mathbb{Z}_q^m|}{|\ker(A)|} = \frac{q^m}{|\ker(A)|}$$

Notice that for every subgroup $K \subseteq \mathbb{Z}_q^m$, the dot product on $\mathbb{Z}_q^m$ is nondegenerate, therefore

$$|K^{\perp}| = \frac{q^m}{|K|}$$

so, applied to $\ker(A)$,

$$|\ker(A)^{\perp}| = \frac{q^m}{|\ker(A)|} = |im(A)|$$

which, by transposition properties, $|im(A)| = |im(A^T)|$, thus

$$|\ker(A)^{\perp}| = |im(A^T)|$$

hence,

$$(\ker(A)^{\perp}) = im(A^T)$$

Continuing from $qy \pmod q \in (ker(A))^{\perp}$, we've just seen that $(ker(A))^{\perp} = im(A^T)$, thus

$$\exists s \in \mathbb{Z}^n \text{ s.th. } qy = A^T s \pmod q$$

Therefore, for some

$$z \in \mathbb{Z}^m,\ qy = A^T s + qz$$

which means that $qy \in \mathcal{L}$, by definition of $\mathcal{L}$ (1.11). Divide that by q, obtaining

$$y \in \frac{1}{q}\mathcal{L}$$

Thus, $\forall y \in \mathcal{L}^{\perp *}$, we have $y \in \frac{1}{q}\mathcal{L}$.

Hence $\mathcal{L}^{\perp *} \subseteq \frac{1}{q}\mathcal{L}$.

Putting together (i). $\mathcal{L}^{\perp *} \supseteq \frac{1}{q}\mathcal{L}$, and (ii). $\mathcal{L}^{\perp *} \subseteq \frac{1}{q}\mathcal{L}$, we have

$$\mathcal{L}^{\perp *} = \frac{1}{q}\mathcal{L}$$

□

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